

# UNIT 6: REPLACEMENT THEORY (6 Hrs.)

Introduction, Replacement of capital equipment which depreciated with time, replacement by alternative equipment, Group and individual replacement policy

## # Past Questions

**2021 -11a.** A Scooter with first cost Rs. 80,000 has depreciation and service pattern as given below:

| Item                     | 1     | 2     | 3     | 4     | 5     | 6    |
|--------------------------|-------|-------|-------|-------|-------|------|
| Depreciation during year | 28000 | 20000 | 14000 | 5000  | 4000  | 4000 |
| Annual Service Charge    | 18000 | 21000 | 25000 | 29000 | 34000 | 4000 |

How many years should the scooter be kept in the service before replacement ?

**2023 – 6.** Find the cost per period of individual-replacement policy of an installation of 3000 street lamps given in the following table.

- a. Cost of replacing an individual bulb is rupees 5.
- b. Cumulative probability of failure is given below

|                                   |   |     |     |     |     |     |
|-----------------------------------|---|-----|-----|-----|-----|-----|
| Week                              | 0 | 1   | 2   | 3   | 4   | 5   |
| Cumulative probability of failure | 0 | 0.1 | 0.3 | 0.5 | 0.7 | 1.0 |

## Introduction

In fact, in any system the efficiency of an item deteriorates with time. In such cases, either the old item should be replaced by a new item, or some kind of restorative action (maintenance) is necessary to restore the efficiency of the whole system. The cost of maintenance depends upon a number of factors, and a stage comes at which the maintenance cost is so large that it is more profitable to replace the old item. Thus, there is a need to formulate the most effective replacement policy.

**Replacement Theory** is used in Operations Research to determine the **optimum time** to replace an asset (like machinery, equipment, vehicles, bulbs, etc.) so that the **total cost is minimized** and **productivity is maintained**.

### Reasons for Replacement:

- Equipment becomes **obsolete** or **inefficient**.
- **Repair and maintenance costs increase** with time.
- New equipment is available that is **cheaper, faster, and more efficient**.
- **Sudden failures** in items like bulbs, fuses, etc.

### Replacement models:

The various types of replacement problems can be broadly classified in following categories:

1. **Replacement of items whose efficiency deteriorates with time / maintenance cost increase with time, (e.g., machine, tools, etc.)**

Items such as machines, vehicles, tires, etc. whose efficiency deteriorates with age due to constant use and which need increased operating and maintenance cost.

In such cases deterioration level is predicted and is represented by:

- (a) Increased maintenance/operation cost.
- (b) Decreasing salvage value i.e. its waste or scrap value and damage to them and safety risk.

**Assumption:** Increased age reduces efficiency

Generally, the criterion for measuring efficiency is the discounted value of all future costs associated with each policy.

## 2. Replacement of items that fail suddenly and completely like electric bulbs & tubes.

In some situations, failure of a certain item occurs all of a sudden, instead of gradual deterioration (e.g., failure of light bulbs, tubes, etc.). The failure of the item may result in complete breakdown of the system. The breakdown implies loss of production, idle inventory, idle labor, etc. Therefore, an organization must prepare itself against these failures.

Thus, to avoid the possibility of a complete breakdown, it is desirable to formulate a suitable replacement policy.

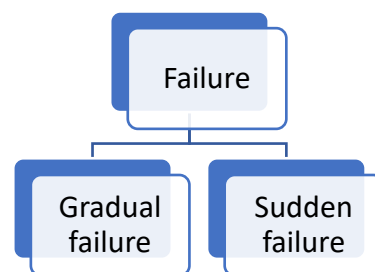
The following two courses can be followed in such situations.

- **Individual replacement policy.** Under this policy, an item may be replaced immediately after its failure.
- **Group replacement policy.** Under this policy, the items are replaced in group after a certain period, say  $t$ , irrespective of the fact that items have failed or not. If any item fails before its preventive replacement is due, then individual replacement policy is used.

In situations where the items fail completely, the formulation of replacement policy depends upon the probability of failure. Mortality tables or Life testing techniques may be used to obtain a probability distribution of the failure of items in a system.

### Types of Failure

There are two types of failure are discussed below;



**Gradual Failure:** It is a progressive in nature. That is, as the life of an item increases, its operational efficiency also deteriorates resulting in

- a. Increased the running cost.
- b. Decreased in its productivity
- c. Decreased in the resale or salvage value

Mechanical items like pistons, rings, bearing tyres etc. fall under this category.

**Sudden Failure:** This type of failure occurs in terms after some period of giving desired service rather than deterioration while in service. The period of giving desired services is not constant but follows some frequency distribution which may be *progressive, retrogressive or random in nature*.

# Replacement of Capital Equipment That Depreciates Over Time

Some assets **wear out gradually** due to usage and aging. Their:

- **Maintenance cost increases**
- **Efficiency decreases**
- **Resale value (salvage value) decreases**

**Objective:** To find the **best time (year)** to replace such equipment **before the total cost becomes too high**.

## Steps to Determine Optimal Replacement Time:

1. List:
  - Purchase cost
  - Maintenance costs for each year
  - Resale/salvage value at end of each year
2. Calculate:
  - **Total Cost (TC)** = Purchase Price + Cumulative Maintenance Cost – Resale Value
  - **Average Annual Cost (AAC)** = TC / No. of years
3. **Replace the equipment** at the year where **AAC is minimum**

**Example 1** A milk plant is considering replacement of a machine whose cost price is Rs. 12,200 and the scrap value Rs. 200. The running (maintenance and operating) costs in Rs. are found from experience to be as follows:  
When should the machine be replaced?

|               |     |     |     |      |      |      |      |      |
|---------------|-----|-----|-----|------|------|------|------|------|
| Year:         | 1   | 2   | 3   | 4    | 5    | 6    | 7    | 8    |
| Running Cost: | 200 | 500 | 800 | 1200 | 1800 | 2500 | 3200 | 4000 |

### Calculations for average cost of machine(In Rupees)

| Year<br>(1) | Running<br>Cost<br>(2) | Cumulative<br>Running<br>Cost<br>(3) | Depreciation<br>Cost<br>(4) | Total Cost TC<br>(5) = (3) + (4) | Average<br>Cost<br>(6) = (5)/(1) |
|-------------|------------------------|--------------------------------------|-----------------------------|----------------------------------|----------------------------------|
| 1           | 200                    | 200                                  | 12000                       | 12200                            | 12200                            |
| 2           | 500                    | 700                                  | 12000                       | 12700                            | 6350                             |
| 3           | 800                    | 1500                                 | 12000                       | 13500                            | 4500                             |
| 4           | 1200                   | 2700                                 | 12000                       | 14700                            | 3675                             |
| 5           | 1800                   | 4500                                 | 12000                       | 16500                            | 3300                             |
| 6           | 2500                   | 7000                                 | 12000                       | 19000                            | <b>3167</b>                      |
| 7           | 3200                   | 10200                                | 12000                       | 22200                            | 3171                             |
| 8           | 4000                   | 14200                                | 12000                       | 26200                            | 3275                             |

From the table it is noted that the average total cost per year is minimum in the 6<sup>th</sup> year (Rs. 3167).

Also the average cost in 7<sup>th</sup> year (Rs.3171) is more than the cost in 6<sup>th</sup> year.

**Hence the machine should be replaced after every 6 years.**

### Example 2

A Machine owner finds from his past records that the maintenance costs per year of a machine whose purchase price is Rs. 8000 are as given below:

Determine at which time it is profitable to replace the machine.

|                                       |      |      |      |      |      |      |      |      |
|---------------------------------------|------|------|------|------|------|------|------|------|
| Year:                                 | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
| Maintenance Cost:                     | 1000 | 1300 | 1700 | 2200 | 2900 | 3800 | 4800 | 6000 |
| Resale Price <b>or</b><br>Scrap Value | 4000 | 2000 | 1200 | 600  | 500  | 400  | 400  | 400  |

| Year<br>(1) | Maintenance<br>cost<br>(2) | Cumulative<br>maintenance<br>cost (3) | Resale<br>Price(4) | Deprecia<br>tion Cost<br>(5) | Total<br>cost<br>(6)=(3)+<br>(5) | Average<br>Cost<br>(7)=(6)/(1) |
|-------------|----------------------------|---------------------------------------|--------------------|------------------------------|----------------------------------|--------------------------------|
| 1           | 1000                       | 1000                                  | 4000               | 4000                         | 5000                             | 5000                           |
| 2           | 1300                       | 2300                                  | 2000               | 6000                         | 8300                             | 4150                           |
| 3           | 1700                       | 4000                                  | 1200               | 6800                         | 10800                            | 3600                           |
| 4           | 2200                       | 6200                                  | 600                | 7400                         | 13600                            | 3400                           |
| 5           | 2900                       | 9100                                  | 500                | 7500                         | 16600                            | <b>3200</b>                    |
| 6           | 3800                       | 12900                                 | 400                | 7600                         | 20500                            | 3417                           |
| 7           | 4800                       | 17700                                 | 400                | 7600                         | 25300                            | 3614                           |
| 8           | 6000                       | 23700                                 | 400                | 7600                         | 31300                            | 3913                           |

The above table shows that the value of Average cost during fifth year is minimum.

**Hence the machine should be replaced after every fifth year.**

## 3. Replacement by Alternative Equipment

Sometimes, better or modern equipment becomes available in the market that can perform the same job with:

- Lower cost
- Higher speed
- Lower energy consumption
- Lower maintenance cost

### Decision Criteria:

Choose the equipment with **lower total cost of ownership** over its useful life.

### How to Compare?

1. Calculate **Total Annual Equivalent Cost** for:
  - Old equipment
  - New alternative
2. Compare costs and **choose the one with the lowest cost**.

**Example:**

- **Old Machine:** Initial cost Rs. 20,000, annual maintenance Rs. 3,000
- **New Machine:** Cost Rs. 35,000, annual maintenance Rs. 1,000
- Useful life of both: 5 years

Compare total costs:

- Old:  $20,000 + (3,000 \times 5) = 35,000$
- New:  $35,000 + (1,000 \times 5) = 40,000$

So, Old is still cheaper. But if new saves production time, that benefit may justify the cost.

#### 4. Group and Individual Replacement Policy

This is used for items like bulbs, tubes, fuses, batteries, etc., which **fail suddenly**, not gradually.

**a. Individual Replacement Policy:**

- Replace item **only when it fails**
- Best when failure rate is **low**
- Simple but may be **costly** if many items fail frequently

**b. Group Replacement Policy:**

- Replace **all items together at fixed intervals**
- Replace failed items **individually** before group replacement
- Effective when failure **increases over time**

**Example:**

For a certain type of light bulbs( 1000 Nos.), following mortality rates have been observed:



| Weeks                                                   | 1  | 2  | 3  | 4  | 5   |
|---------------------------------------------------------|----|----|----|----|-----|
| Percentage of items failing by at the end of each month | 10 | 25 | 50 | 80 | 100 |

Each bulb costs Rs.10 to replace an individual bulb on failure. If all bulbs were replaced at the same time in group it would cost Rs. 4 per bulb. It is under proposal to replace all bulbs at fixed intervals of time, whether or not the bulbs have burnt out. And also it is to continue replacing immediately burnt out bulbs.

**Determine the time interval at which all the bulbs should be replaced?**



### SOLUTION:

Let  $p_i$  = the probability that a new light bulb fails during the  $i$ th week of its life.

Thus

$p_1$  = the probability of failure in 1st week =  $10/100 = 0.10$

$p_2$  = the probability of failure in 2nd week =  $(25-10)/100 = 0.15$

$p_3$  = the probability of failure in 3rd Week =  $(50-25)/100 = 0.25$

$p_4$  = the probability of failure in 4th week =  $(80-50)/100 = 0.3$

$p_5$  = the probability of failure in 5th week =  $(100-80)/100 = 0.2$

Since the sum of all the above probabilities is unity, the further probabilities  $p_6$ ,  $p_7$ ,  $p_8$  and so on, will be zero.

Thus, all light bulbs are sure to burnout by the 5th week.

Furthermore, it is assumed that bulbs that fail during a week are replaced just before the end of that week.

Let  **$N_i$  = the number of replacements made at the end of the  $i$ th week.**

And let all 1000 bulbs are new initially.

Thus,

$N_0$  = the number of replacements made in the beginning.

$N_1$  = the number of replacements made at the end of the 1<sup>st</sup> week

$N_2$  = the number of replacements made at the end of the 2<sup>nd</sup> week

$$N_0 = N_0 = 1000$$

$$N_1 = N_0 p_1 = 1000 \times 0.10 = 100$$

$$N_2 = N_0 p_2 + N_1 p_1 = 1000 \times 0.15 + 100 \times 0.10 = 160$$

$$N_3 = N_0 p_3 + N_1 p_2 + N_2 p_1 = 1000 \times 0.25 + 100 \times 0.15 + 160 \times 0.10 = 281$$

$$N_4 = N_0 p_4 + N_1 p_3 + N_2 p_2 + N_3 p_1 = 377$$

$$N_5 = N_0 p_5 + N_1 p_4 + N_2 p_3 + N_3 p_2 + N_4 p_1 = 350$$

$$N_6 = 0 + N_1 p_5 + N_2 p_4 + N_3 p_3 + N_4 p_2 + N_5 p_1 = 230$$

$$N_7 = 0 + 0 + N_2 p_5 + N_3 p_4 + N_4 p_3 + N_5 p_2 + N_6 p_1 = 286$$

### Individual replacement:

The mean age of bulbs =  $1 \times p_1 + 2 \times p_2 + 3 \times p_3 + 4 \times p_4 + 5 \times p_5$

$$= 1 \times 0.1 + 2 \times 0.15 + 3 \times 0.25 + 4 \times 0.30 + 5 \times 0.20 = 3.35 \text{ Weeks,}$$

The number of failures in each week in steady state =  $1000/3.35 = 299$

And the cost of replacing bulbs individually on failure

$$= 10 \times 299 \quad (\text{at the rate of Rs. 10 per bulb})$$

$$= \text{Rs. } 2990$$

### Group replacement calculation:

The replacement of all 1000 bulbs at the same time in bulk costs Rs. 4 per bulb and replacement of an individual bulb on failure costs Rs. 10. Costs of replacement of all bulbs simultaneously are calculated in the Table

| End of week | Cost of individual replacement | Total cost of group replacement (Rs.)  | Average cost per week (Rs.) |
|-------------|--------------------------------|----------------------------------------|-----------------------------|
| 1           | $100 \times 10 = 1000$         | $1000 \times 4 + 100 \times 10 = 5000$ | 5000.00                     |
| 2           | $160 \times 10 = 1600$         | $5000 + 160 \times 10 = 6600$          | 3300.00                     |
| 3           | $281 \times 10 = 2810$         | $6600 + 281 \times 10 = 9410$          | 3136.67                     |
| 4           | $377 \times 10 = 3770$         | $9410 + 377 \times 10 = 13180$         | 3295.00                     |

**Table** Cost of replacement of bulbs for Bulbs example

The cost of individual replacement in the 4th week is greater than the average cost for 3 weeks.

Therefore the **optimal replacement decision is to replace all the bulbs at the end of every 3 weeks.**

**Answer**

### # Solution of Past Questions

**2021 -11a.** A Scooter with first cost Rs. 80,000 has depreciation and service pattern as given below:

| Item                     | 1     | 2     | 3     | 4     | 5     | 6    |
|--------------------------|-------|-------|-------|-------|-------|------|
| Depreciation during year | 28000 | 20000 | 14000 | 5000  | 4000  | 4000 |
| Annual Service Charge    | 18000 | 21000 | 25000 | 29000 | 34000 | 4000 |

How many years should the scooter be kept in the service before replacement ?

**Solution:**

| Year | Running Cost (2) | Cumulative Running Cost (3) | Depreciation Cost (4) | Total Cost = (3)+(4) | Average Cost = (5)/(1) |
|------|------------------|-----------------------------|-----------------------|----------------------|------------------------|
| 1    | 18,000           | 18,000                      | 28,000                | 46,000               | 46,000                 |
| 2    | 21,000           | 39,000                      | 20,000                | 59,000               | 29,500                 |
| 3    | 25,000           | 64,000                      | 14,000                | 78,000               | 26,000                 |
| 4    | 29,000           | 93,000                      | 5,000                 | 98,000               | 24,500                 |
| 5    | 34,000           | 127,000                     | 4,000                 | 131,000              | <b>26,200</b>          |
| 6    | 40,000           | 167,000                     | 4,000                 | 171,000              | 28,500                 |

From the table we can observe that the **minimum average cost is Rs. 24,500**, which occurs at the **end of year 4**.

**Thus, the scooter should be kept for 4 years before replacement.**

**2023 – 6.** Find the cost per period of individual-replacement policy of an installation of 3000 street lamps given in the following table.

- Cost of replacing an individual bulb is rupees 5.
- Cumulative probability of failure is given below

|                                   |   |     |     |     |     |     |
|-----------------------------------|---|-----|-----|-----|-----|-----|
| Week                              | 0 | 1   | 2   | 3   | 4   | 5   |
| Cumulative probability of failure | 0 | 0.1 | 0.3 | 0.5 | 0.7 | 1.0 |

Soln:

Let,  $P_i$  be the probability that a lamp bulb fails during the  $i^{\text{th}}$  week of its life. Then,

$$P_1 = 0.1$$

$$P_4 = 0.7 - 0.5 = 0.2$$

$$P_2 = 0.3 - 0.1 = 0.2$$

$$P_5 = 1.0 - 0.7 = 0.3$$

$$P_3 = 0.5 - 0.3 = 0.2$$

Since, the sum of all above probabilities is 1. So, the probabilities greater than  $P_5$  are zero

Thus, all lamp bulbs are sure to burn out by 5<sup>th</sup> week.

Let,  $N_i$  be the no. of replacement made at the end of  $i^{\text{th}}$  week

Thus,  $N_0 = 3000$  (Initial no. of bulb)

$$N_1 = N_0 P_1 = 3000 \times 0.1 = 300$$

$$N_2 = N_0 P_2 + N_1 P_1 = 3000 \times 0.2 + 300 \times 0.1 = 630$$

$$N_3 = N_0 P_3 + N_1 P_2 + N_2 P_1 = 3000 \times 0.2 + 300 \times 0.2 + 630 \times 0.1 = 723$$

$$N_4 = N_0 P_4 + N_1 P_3 + N_2 P_2 + N_3 P_1 = 858.3$$

$$N_5 = N_0 P_5 + N_1 P_4 + N_2 P_3 + N_3 P_2 + N_4 P_1 = 1316.43$$

Now,

$$\text{Expected life of each bulb} = \sum_{i=1}^5 i P_i$$

$$= 1 \times 0.1 + 2 \times 0.2 + 3 \times 0.2 + 4 \times 0.2 + 5 \times 0.3$$

$$= 3.4$$

$$\therefore \text{Average no. of failure per week} = \frac{3000}{3.4} = 882 \text{ (approx)}$$

$$\therefore \text{cost of individual replacement} = 882 \times \text{Rs. 5} \\ = \text{Rs. 4410} //$$